

Solution Requirements

Date	15 March 2026
Team ID	XXXXXX
Project Name	RK Health – AI Smart Patient Appointment & Medication Reminder System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely register and log in using a username/email and password. The system validates credentials before granting dashboard access.	High
2	Authorization levels	<i>[The system supports Patient and Admin roles. Patients manage appointments and medications, while Admins</i>	High

		<i>monitor records and system activities.]</i>	
3	External interfaces	Integrates with Google Apps Script, Google Sheets, Groq AI API, Twilio SMS API, and Google Calendar API for healthcare management.	High
4	Transactions processing	The system processes appointment booking, medication scheduling, AI summary generation, SMS reminders, calendar events, and health report generation. All records are stored in Google Sheets.	High
5	Reporting	Generates healthcare dashboards, medication history, appointment reports, AI-generated visit summaries, and downloadable PDF health reports.	High
6	Business rules, etc.	Appointments require valid patient details and future dates. Medication reminders require complete medicine information. AI summaries are generated only after a patient visit record is available.	High

7	Compliance to laws or regulations	Patient health data is securely stored. API keys are protected. Basic privacy and secure data handling practices are followed.	Medium
8	<i>Other</i>	Supports wearable device integration, predictive healthcare recommendations, advanced analytics, responsive design, and secure cloud-based record management.	Medium

S.No	NFR Category		Target Metric / Acceptance Criteria
1	Performance & Speed	Dashboard, reports, and AI summary generation should be responsive with minimal waiting time.	Dashboard loads in less than 2 seconds ; AI summary generated in less than 10 seconds .
2	Scalability	The system should support increasing numbers of users and healthcare records without major performance degradation.	Supports 1,000+ users and 50,000+ healthcare records .
3	Security & Data Privacy	Protect patient information using secure authentication, HTTPS communication, protected API keys, and restricted backend access.	HTTPS enabled, API keys stored securely, authenticated access only.
4	Reliability & Availability	The application should remain available with reliable cloud services and automatic data storage.	99.9% uptime with reliable Google Cloud services.

5	Usability & Accessibility	The interface should be simple, responsive, mobile-friendly, and accessible across modern browsers and devices.	Fully responsive and compatible with Chrome, Edge, Firefox, and mobile browsers.
6	<i>Other</i>	Maintainable, portable, modular, and easy to deploy using GitHub Pages and Google Apps Script. Future-ready for wearable integration and predictive AI features.	Easy deployment, modular architecture, and future scalability.